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COMMUNICATION

Accessibility, usability, and security evaluation of universities' prospective student web pages: a comparative study of Europe, North America, and Oceania

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Abstract
Universities' prospective student web pages aim to disseminate information about their academic and social opportunities to their stakeholders; therefore, they must be accessible, of high quality of use and reliable. This article presents the accessibility, usage performance, and security analysis of prospective student web pages of 330 universities from three continents, namely Europe, North America, and Oceania. For this purpose, university websites were selected based on the Webometrics ranking, and online automated test tools were used. The results showed that websites at universities in North America paid more attention to accessibility and quality of use on prospective student web pages, followed by Oceanian and European websites. Evaluated websites had low compliance levels according to the WCAG 2.0 guideline. No major problems were identified in terms of usability and security, but there were certain points for improvement. Moreover, we present and discuss recommendations to developers and administrators for websites to resolve accessibility, usability, and security breaches and provide information equally to all stakeholders. Hence, this analysis report provides feedback to web developers to improve accessibility, quality of use, and security issues of university websites and their prospective student web pages.

Keywords Human–computer interaction · Web information systems · Web usability · Web accessibility · Web security · Web performance evaluation

1 Introduction

With the rapid spread of the use of Internet, the promotion of the services offered has also been facilitated through websites and having a website has become essential for institutions and organizations. Websites also play an important role for universities. The widespread use of e-learning, open learning, and distance learning also makes it a necessity for a university to have a website [1]. University web pages allow universities to announce their opportunities to the world and increases their competitive power around the world as well. University websites not only provide information for current students, but also assist in attracting prospective students. The corresponding pages of university websites help students in a university and a faculty selection. Moogan [2] stated that university websites are the most frequently referenced source of information after the university prospectus. In an another study, 94% of prospective students stated that they would examine the website before considering a university [3]. Moreover, college websites have become a marketing tool targeting prospective students [4].

University websites contain various pages that present different information to their stakeholders. There are special pages that provide information for prospective students who have a wide audience among the stakeholders. Prospective student pages provide general information about the university, academic information, academic units, campus, and application conditions. Prospective student links provide a user-oriented home page in which students can use the necessary information and terminology, creating a welcoming environment for students [4].

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With the popularity of website evaluation, many researchers have conducted studies in different areas to improve website quality. In this manner, widely used application areas include e-commerce, government, education, and medical sectors [5–23]. The criteria for evaluating a website can be: usability, authority, performance, traffic rank, link popularity, functionality, accessibility, security, design patterns, HTML syntax analysis, and browser compatibility [24]. These evaluations can be made by expert assessment, automated tools, or user experience methods. Automated testing tools are a quick and easy way to test the quality of websites. There are many available automated testing tools for this purpose. Using such tools, it is possible to evaluate accessibility of websites in a time and cost-effective manner. The quality levels of the websites reviewed in the literature indicate the need for a tool-based approach to support manual testing.

Rowland [25] emphasizes the importance of access to education for students with disabilities, especially via the Internet. Thus, accessibility is a key factor for university websites. The Web Content Accessibility Guidelines (WCAG) produced by the World Wide Web Consortium (W3C) have been adopted globally by higher education institutions to ensure accessibility [26]. The level of individuals accessing, understanding, interacting, and navigating a website can be measured with accessibility analysis. Similarly, usability is an important issue for university websites and their quality. It is necessary to pay attention to the application of certain design principles on university websites. Usability inspections test the loading speed of the home page, page size, and number of broken links [27, 28]. Another important criterion for websites is security; it increases the risk of the site being attacked if the system is not up to date and the communication is not made over an encrypted channel. Websites that have become vulnerable to attacks such as SQL injection and cross-site scripting may cause service interruptions and data loss, or become completely dysfunctional [17].

It is clear from a recent literature review [29] that most of the studies in this field have evaluated universities' home pages, while there has been no work investigating the prospective student web pages as well. Further, according to this review [29], the related studies on university web pages from Europe are quite low, and very few works have compared the performances of university web pages from different countries. In view of these research gaps, the purpose of this study is to comprehensively assess the prospective student web pages of universities from three continents, namely, Europe, North America, and Oceania, in terms of accessibility, usability, and security. This study attempts to eliminate gaps in the literature by analyzing prospective student pages with automated test tools according to these three basic criteria. In addition, the main accessibility and usability problems on prospective student web pages are determined according to WCAG 2.0. Finally, the study presents recommendations for improving the quality of prospective student web pages by identifying accessibility, usability, and security issues.

The rest of this study is organized as follows: in Sect. 2, a literature review is presented that includes studies conducted by different researchers in terms of accessibility, usability, and security of websites, and reviews of university websites. The definition of the data used within the scope of the study, the tools used for the analysis of these websites, and the evaluation process are explained in Sect. 3. We provide the results obtained as a result of accessibility, usability, and security analyses in Sect. 4. In Sect. 5, suggestions and discussion points are given for solving web accessibility, usability, and security issues. In the last section, the study concludes and possible directions for future studies are presented.

2 Literature review

2.1 Web accessibility

Accessibility evaluation of websites is carried out to verify whether all individuals (including people with disabilities) are equal in accessing information [30]. Most of the accessibility analyses in current studies used the WCAG 2.0 guide published by the World Wide Web Consortium (W3C) in 2008 [31]. WCAG is designed to be widely applied to different web technologies now and in the future and can be tested with a combination of automated testing tools and human evaluation [32]. Within the scope of this study, websites were analyzed using automated evaluation tools. There are many different web accessibility evaluation tools [33]. Automated web accessibility evaluation tools are used to identify potential accessibility issues, and these tools are also useful for manual review [34]. Website accessibility analysis using automatic evaluation tools has been discussed in many studies.

In one such study, researchers compared the accessibility of university websites in India with top university websites in different countries [35]. While HERA was used for manual testing, the Firefox Accessibility Evaluation Toolbar and the TAW tool were used for automated testing. Although most countries have accessibility guidelines for educational institutions, the results show that only 50% of websites comply with these guidelines. Ismailova and Inal [6] examined government websites in Kyrgyzstan, Azerbaijan, Kazakhstan, and Turkey for accessibility and quality in use. Accessibility analysis of 16 websites for each country was performed using the AChecker online tool [36]. The accessibility analysis results showed that all four countries

violated many checkpoints and the least accessibility error occurred on Kazakh government websites. Tashtoush et al. [37] conducted accessibility testing of Arabic and English versions of government websites randomly, selected from ten countries. These accessibility tests were performed using the AChecker, TAW [38], WAVE [39], and Sort Site [40] tools, using the same procedures for both the English and Arabic versions. Researchers examined accessibility issues of 20 government websites in Bangladesh [18]. The results showed that Bangladeshi government websites contain a high rate of accessibility errors. The research was performed by using various tools like TAW, WAVE, CCA, Total Validator, CynthiaSays, Keyboard Accessibility, and Readability test.

Kaur et al. [15] aimed to analyze the accessibility of the websites of 280 hospitals in major cities of India. Accessibility tests were performed only with the online test tool TAW. As a result of the evaluations, a total of 22,491 errors were found, which meant an average of 80.32 errors per website. Accessibility, usability, and security assessment were discussed in a study on banking websites in Turkey [41]. A total of 32 banking websites were analyzed according to the WCAG 2.0 guidelines criteria. The results showed that the websites evaluated had significant accessibility, usability, and vulnerability issues.

2.2 Web usability

ISO 9241–11 interprets usability as "the extent to which the system, product or service can be used and satisfied by specified users to achieve certain goals" [42]. In one study, McCloskey revealed a significant and positive relationship between trust and ease of use [43]. The importance of usability in the study of online behaviors has been emphasized in human–computer interaction (HCI) and information systems (IS) research [44]. The literature presents studies evaluating the usability of websites operating in areas such as education, social media, and government.

Youngblood and Mackiewicz [45] investigated the relationship between availability rates of municipal government websites in Alabama, USA, and population and per capita income. W3C's online HTML Validation Service and the WAVE tool were used for evaluation. Karaim and Inal [19] analyzed the Management of Scholarships website based on Nielsen's heuristic assessment. The results showed that the website had many usability problems. Kaur et al. [46] evaluated the websites of universities in Punjab using four automatic evaluation tools (Pingdom, GTmetrix, Website Grader, and Site Speed Checker) and presented comparative results based on various factors. Zarish et al. [47] created a comprehensive study by evaluating the websites from different aspects such as Accessibility, Content, Design, Mobile Readiness, Page Analysis, Performance, Search Engine Optimization, Security and Usability. In the study, the researchers examined Pakistan's educational websites using three automated tools (Qualidator, Website Grader, and Website Analyzer) and ranked the sites according to their scores as a result of the analysis. In another study [48], usability analysis of seven social media websites was performed with three different automated evaluation tools Pingdom, GTmetrix, and Website Grader.

2.3 Web security

Ratnasingam argues that security is actually a big problem and points out the potential for a hacker to hijack or destroy an application [49]. In this context, related studies in the literature including security analysis of websites were examined.

Dukes et al. [50] compared automated tools and manual testing methods for vulnerability detection. The result of the evaluation showed that some problems can be detected by expert observation and determined manually, but more errors can be detected by using these two methods together. Alarifi et al. [51] evaluate the security of the most visited and popular Arab websites. In the security analysis, the status of carrying malware, being on the black list and having up-to-date software, is investigated in 7000 different domains. Researchers use APIs such as Sucuri SiteCheck, McAfee SiteAdvisor, Google Safe Browsing, and Norton offered by many well-known web browsers to detect suspicious situations.

2.4 Related work on universities websites

The web is probably the main showcase for universities, but in the near future the virtual institution may be as important and representative as a real institution [52]. The fact that higher education institutions have many stakeholders also supports the mentioned prediction. University web pages present various information about academic programs, course contents, student affairs, teaching facilities, campus facilities, etc. [53]. Higher education institutions' websites have an important role in providing information to prospective students, current students, faculty, and graduates [54]. There are various studies in the literature in which websites of higher education institutions are evaluated according to different criteria.

Patricia Acosta-Vargas et al. [7] evaluated the web sites of 20 universities in North America, Latin America, Europe, Asia, Africa, and Oceania ranked by Webometrics for accessibility. As a result of the evaluation, it was seen that even the websites of universities with high academic prestige did not reach an acceptable level of compliance. The results of an another study [55] examining the accessibility problems of websites of 348 universities in Latin America selected

randomly from Webometrics showed that there were many problems related to website accessibility. Researchers at Griffith and Princess Nora University conducted accessibility assessments of 20 universities and 180 websites. Three pages of each university, the home page, the admission page, and a course description page were tested with the AChecker automated evaluation tool, and the results were compared with the studies conducted between 2005 and 2015. Comparison of the reviews in the literature showed that the issue of accessibility continues to attract the attention of universities, educational institutions, developers, and professors [56]. In another study, researchers examined the accessibility of websites of Higher Education Institutions in Cyprus. Three automated tools (TAW, EIII page Checker, and WAVE) were used against the WCAG 2.0 standard. Findings showed that no one passed the test without any errors. Many of the websites have reached low availability levels [57]. Ringlaben et al. [58] evaluated special education department websites in the USA. A total of 51 websites were tested using the AChecker and Bobby tools for evaluation. While almost all of the websites examined were found to have accessibility problems, it was stated that 39% of the websites had problems that need to be resolved urgently. Nir and Immerman [59] conducted a study that included the websites of higher education institutions in Israel. The findings showed that all web pages examined had accessibility problems. In another study that evaluated home pages of universities in Spain, Chile, and Mexico, researchers analyzed the top 15 universities in the Webometrics rankings [8]. Verkijika and De Wet [9] aimed to examine the accessibility issues of 26 university websites in South Africa. As a result of the analysis, it was found that all of the websites were incompatible with WCAG 2.0. The most common error identified is a broken connection problem, and 4 of the websites failed the Google Mobile-friendly Test.

Kaur et al. [60] used two automated tools, Site Analyzer and Qualidator, to assess the usability levels of 10 Education Universities in the Punjab region. Researchers present rating and usability rates according to the evaluation criteria for each university as a result of the evaluation. Ismailova and Kimsanova [28] evaluated the accessibility, usability, and security aspects of 42 university websites in Kyrgyzstan using the EvalAccess automatic evaluation tool. These websites have been found to violate many checkpoints in the WCAG 1.0 guide. In addition, it has been observed that websites give a low priority to the security problem. It is stated that most of the websites that use content management systems (corresponding to 81% of the websites) are outdated and therefore have a security vulnerability.

Regarding Turkey, Yerlikaya and Durdu [61] examined the accessibility problems of 20 university websites, randomly selected based on URAP (University Ranking by Academic Performance) scores. Analysis of findings shows that there is no university website that meets the accessibility criteria. Similarly, Akgül [62] has reported similar problems examining the websites of public and private universities in Turkey. Researchers have stated that private university websites are better than state university websites but emphasized that more resources should be allocated for accessibility problems. Kurt [63] examined the availability of 10 randomly selected university websites. Websites were tested using the Web Accessibility Checker, AChecker, and SortSite automated testing tools. As a result of the analysis, the researcher found all websites to have accessibility problems. In the following years, in an initial study, the author observed the development of websites using the same data [11]. Compared to the author's previous study, it was observed that the accessibility problems of websites were reduced.

Considering the gap in the literature, in this study, the status of university web pages developed for prospective students is evaluated in a unique context. The prospective student pages of universities may be an example of the main showcase described by Aguillo et al. [52]. The commercial interests of these pages, which are primarily intended for prospective students and parents, are an undeniable fact. In this context, this study was conducted to better understand the accessibility, usability, and security problems of web pages and to guide the developers.

3 Methodology

3.1 Data

This study evaluated the prospective students' web pages of universities from three continents, Europe, North America, and Oceania, in terms of accessibility, usability, and security. The rankings in the "Webometrics Ranking of World Universities" developed by the Cybermetrics Lab were taken into account when choosing the university [64]. The English versions of the websites of the universities included in these rankings were examined, and the top 10 universities with prospective student pages for each country were determined.

Countries that do not have a university in the ranking, do not have enough universities, or do not have a page for prospective students were not included to the study. Thus, 330 universities from 33 countries (10 university pages for each university) were selected and evaluated. The country and Webometrics ranking information of the selected universities are given in Table 1.

3.2 Tools and techniques

Automatic evaluation tools such as TAW were used for web accessibility analysis. Each prospective student page was

Table 1 Countries included in the study and the Webometrics ranking median

Country	Continent	Median of the ranking
USA	North America	8.5
UK	Europe	80
Canada	North America	109
Australia	Oceania	114
Germany	Europe	115.5
Sweden	Europe	131.5
Netherlands	Europe	193
Italy	Europe	212
Spain	Europe	271
France	Europe	321
Finland	Europe	401.5
Belgium	Europe	411.5
Switzerland	Europe	516
Austria	Europe	557
Ireland	Europe	964
Turkey	Europe	1048
Russian Federation	Europe	1094
Portugal	Europe	1148
New Zealand	Oceania	1168
Poland	Europe	1216.5
Czech Republic	Europe	1271
Hungary	Europe	1385
Norway	Europe	1499
Romania	Europe	2724.5
Slovakia	Europe	3474
Bulgaria	Europe	3584.5
Ukraine	Europe	3864.5
Belarus	Europe	4416
Denmark	Europe	5678
Lithuania	Europe	8062.5
Latvia	Europe	9011.5
Serbia	Europe	10,708.5
Croatia	Europe	12,798

evaluated using the test tool according to WCAG 2.0 standards. The summary and details of the report presented by the TAW tool provide an overview based on accessibility guides grouped into perceivable, operable, understandable, and robust categories [65]. In this report, the results of detected situations in each success criterion are presented in terms of the following categories: "Problems found", "No problems found", "A human review is recommended" and "Unable to perform automatic review". Within the scope of the study, only results related to categories of "Problems found" and "No problems found" were considered and discussed. The TAW tool has been used by many researchers such as Akgül and Vatansever [66] and Ismail and Kuppusamy [67], and its effectiveness has shown in related studies.

The GTmetrix online test tool [68] was used for usability testing. This tool allows to calculate the overall speed score and rating of the web page, the loading speed, the total page size, and the total number of requests. The tool was used by Muhammad Bilal, Zhi Yu, Shuyi Song, and Can Wang [16] when testing Pakistani and Chinese government websites to calculate load time and page size. GTmetrix measures the performance of the site in various aspects and presents the performance score from the Google Lighthouse test. Performance score influences the GTmetrix Grade, which provides information on how fast the page is performing. This performance score is calculated based on the page's performance against six key metrics. These six basic metrics are as follows:

- First contentful paint (FCP);
- Speed index (SI);
- Largest contentful paint (LCP);
- Time to interactive (TTI);
- Total blocking time (TBT);
- Cumulative layout shift (CLS);

Each performance criterion generated by the GTmetrix was compared with a certain threshold value, and the result forms the performance score. Within the scope of our study, the fully loaded time, first contentful paint, speed index, and total blocking time metrics were analyzed. Fully Loaded Time specifies the time the web page loads completely, and this time is directly related to six metrics. It cannot be considered a clear performance indicator for users as it waits for the page to complete all data loading and for the JavaScript code to stop executing. First contentful paint (FCP) is a metric that gives information about the time it takes for text, images, etc., on the page to start loading and rendering on the page. Speed index (SI) is a metric that shows how quickly a web page's visible content fills up. SI is directly related to other performance metrics, so it is an appropriate metric for benchmarking page performance. Total blocking time (TBT) is a metric that refers to the amount of time users are blocked from interacting on a web page. This metric accounts for 25% of the performance score and has the highest rate among other metrics.

The Sucuri test tool was used for security testing of websites [69]. According to Csontos et al. [20], this testing tool examines the details of websites in sufficient detail to help determine whether a particular website represents a security risk to users. The Sucuri tool presents the risk level of the website, the status of using TLS (SSL), the status of using the website engine or the programming language it was developed in, the server type and version in the report.

4 Results

4.1 Accessibility results

Website accessibility provides equal opportunities for all users, including people with disabilities, to perceive, understand, navigate, and interact with a website. This section presents the results of the accessibility analysis of prospective student websites using the TAW automated testing tool.

Using the TAW tool, 312 websites (18 websites in the dataset could not be tested due to timeout) were found to have accessibility requirements that were not met in compliance level AA. Considering the error distributions of all evaluated websites, the errors in the robust principle had the highest error rate with a share of about 45%, and the understandable principle had the lowest error rate with 5%. Figure 1 shows the number of accessibility problems per website across the four basic principles, i.e., perceivable, operable, understandable, and robust. Looking at these results, the continent with the minimum number of errors per website in all principles is North America.

Table 2 shows the best and worst performing countries in the three continents when looking at the total number of errors. The country with the lowest number of errors per website was Spain, followed by the UK, Sweden, the USA, and Italy. On the other hand, Denmark, Finland, Portugal, Switzerland, and Belarus had the highest number of accessibility errors per website.

Table 3 provides an overview of the number of checkpoints in the WCAG 2.0 guidelines and shows their distribution by continent. The overall column shows the total number of errors violated for the corresponding checkpoint and the percentage of universities that this checkpoint is violated. Similarly, in the results by continents, the percentage of websites violating the specified checkpoint for the corresponding continent and the number of accessibility errors per website for the relevant continent and checkpoint are presented. The highest rate of accessibility errors was at conformance level A. As an overview of the results, the most frequently violated success criteria were 1.1.1 (Text alternative missing) and 4.1.1 (Parsing), which were violated by more than 85% of websites, followed by 1.3.1—Info and Relationships and 2.4.4—Link Purpose. On the other hand, 2.4.2 Page Title is the least violated checkpoint by websites. This result indicates that there are headings that describe the purpose and subject of prospective student pages. Over 75% of websites met checkpoint 3.1.1—Language of Page. This shows that the language used by default on all pages is understandable by the program.

Table 2 Top and bottom five countries in the ranking of accessibility errors per website

Country	Continent	Number of errors per website
Spain	Europe	18.67
UK	Europe	22.22
Sweden	Europe	23.10
USA	North America	23.30
Italy	Europe	25.13
Belarus	Europe	72.44
Switzerland	Europe	90
Portugal	Europe	96.89
Finland	Europe	113.30
Denmark	Europe	159.60

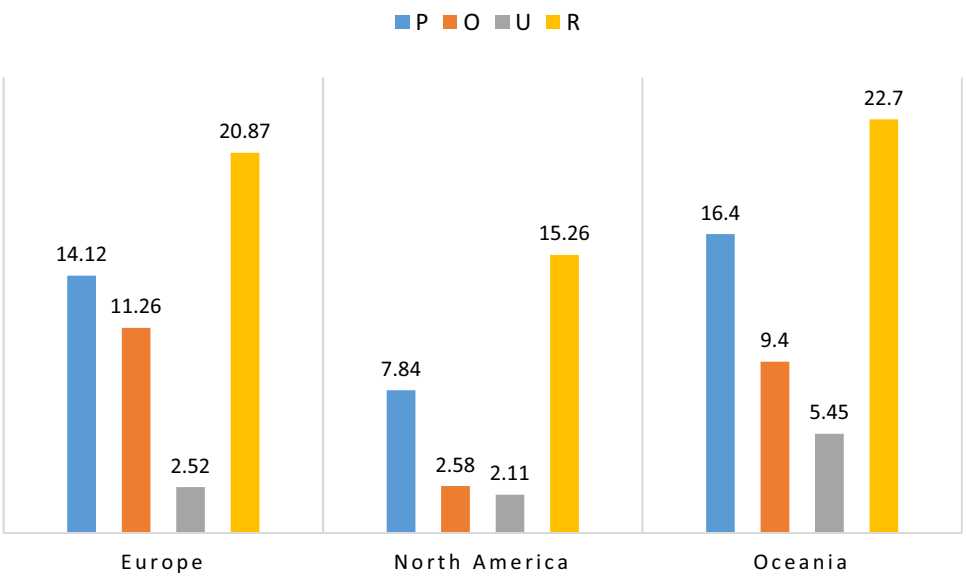


Fig. 1 Accessibility error distributions of the continents across the four basic principles

Table 3 Violations of success criteria at conformance level A

Success criteria	Overall		Continents					
			Europe (n = 273)		North America (n = 19)		Oceania (n = 20)	
	%	Total	%	Avg	%	Avg	%	Avg
1.1.1—Non-text Content	85.58	2173	84.98	8.34	84.21	4.31	95	8.95
1.3.1—Info and Relationships	84.94	2128	85.35	8.11	78.95	5.33	85	8.32
2.4.2—Page Titled	2.24	7	2.20	1	0	0	5	0.05
2.4.4—Link Purpose	81.41	3302	83.88	13.45	68.42	3.77	60	9.05
3.1.1—Language of Page	17.63	69	19.05	1	5.26	1	10	0.84
3.2.2—On Input	23.72	95	21.61	1.27	26.32	1.40	50	0.68
3.3.2—Labels or Instructions	62.18	682	61.17	3.32	68.42	2.46	70	5
4.1.1—Parsing	85.58	5459	84.25	21.30	94.74	13.28	95	16.95
4.1.2—Name, Role, Value	77.24	904	74.36	3.55	94.74	2.83	100	6.95

Table 4 Distribution of GTmetrix grades

GTmetrix Grade (A-F)	Number of websites
A (excellent)	16 (5.11%)
B (good)	34 (10.86%)
C (average)	83 (26.52%)
D (below average)	82 (26.20%)
E (bad)	50 (15.97%)
F (very bad)	48 (15.34%)

All prospective student web pages on the Oceanian continent were found to violate checkpoint 4.1.2—Name, Role, Value. Likewise, 1.1.1—Non-text Content and 4.1.1—Parsing checkpoints were also violated by almost all of the web pages on this continent. On the other hand, the North American continent showed the best performance for the 3.1.1—Language of Page checkpoint.

4.2 Usability results

Page sizes and loading times of 313 prospective student web pages (17 websites in the dataset could not be tested due to timeout and security issues) were analyzed with the online test tool GTmetrix. In the performance grade results, values are determined from A to F, where A corresponds to the best performance grade and F to the worst performance grade.

Table 4 shows the distribution of performance grades. While 5% of the websites evaluated had an A performance score, more than half of the websites reached C and D performance score. Almost a third of web pages had performance grades of E and F.

The percentages of performance grades distribution by continent are presented in Fig. 2. Two-thirds of prospective student web pages in Europe had below-average performance grades (D, E, and F); 45% of prospective student web pages in North America had an A performance score. The

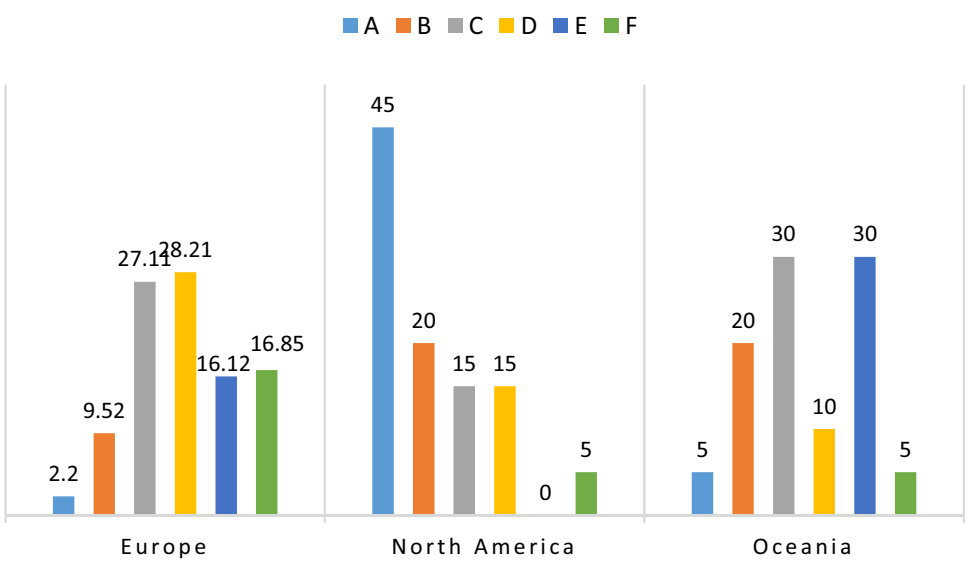


Fig. 2 Percentage of performance grades by continent

rate of websites with E and F performance scores in North America was 5%; in this context, it can be said that the best performance belongs to web pages in North America.

Table 5 contains a summary of the quality performance values of the websites by continent. Fully loaded time of a prospective student web page in Europe exceeded 35 s. Web pages in North America were within an acceptable range for this metric. The list of the top 5 and bottom 5 countries according to the average full load time of the websites is given in Table 6 in detail. Canada and the USA had less than 3 s to load, which was the best performance among all countries. The country with the longest fully loaded time was the UK, followed by Turkey, Hungary, New Zealand, and Russia. The average fully loaded time for these countries was more than 7 s, which was more than the overall average for all continents. Likewise, the mean values of the first contentful paint metric were compared with the threshold values. The results showed that web pages in North America were considered acceptable but could be improved. Web pages in Europe and Oceania were found to take much longer than recommended for this metric. Within the speed index, the average values of web pages in Europe averaged longer than recommended time based on threshold values. The total blocking time metric values showed that at least one website on every continent exceeded the recommended threshold. In general, the blocking time of websites for average values is at an acceptable level.

In the analysis, the total page size of the websites was also discussed. In Table 7, the total size of the websites was analyzed, and the threshold ranges were determined according to the general averages. At least 50% of universities on each continent had a page size of less than 2.4 MB, which is considered generally acceptable performance. Looking at the overall distribution by continents, the results showed that prospective student web pages in European countries had less page size. Almost half of web pages in Oceania have a page size of more than 2.4 MB, which increases the risk of slow loading and clicks.

Table 8 presents a list of the top five countries based on the number of websites in the 0–1.4 MB (very good) and 4.5 MB and above (very bad) ranges. In this context, 80% of the websites in Germany were in this range and had the best performance. This performance was followed by Italy and Romania. On the other hand, three of the websites exceeding 4.5 MB page size belonged to Belarus, which was the maximum number of websites in this range for a country.

Table 5 Quality performance summary of websites by continent

Performance metrics	Continents		
	Europe (n = 273)	North America (n = 20)	Oceania (n = 20)
Fully loaded time (s)			
Min	1.5	0.903	1.5
Mean	5.77	2.82	6.27
Max	35.6	6.4	12.9
First contentful paint (s)			
Min	0.513	0.262	0.619
Mean	2.32	1.03	1.79
Max	10.5	3.1	3.8
Speed index (s)			
Min	1.1	0.419	0.916
Mean	3.01	1.36	2.35
Max	18.3	3.4	5
Total blocking time (ms)			
Min	0	0	0
Mean	56.48	50.05	87.45
Max	814	396	449

Table 6 Ranking of countries by average fully loaded times

Country	Continent	Average of fully loaded time (s)
Canada	North America	2.73
USA	North America	2.91
Bulgaria	Europe	4.13
Czech Republic	Europe	4.31
France	Europe	4.34
Russian Federation	Europe	7.60
New Zealand	Oceania	7.73
Hungary	Europe	8.42
Turkey	Europe	8.55
UK	Europe	12.31

4.3 Security results

Within the scope of the security analysis, the risk levels of the websites were examined. The SUCURI tool groups the risk levels as: minimal, low, medium, high, and critical. More than half of the websites evaluated have a low risk

Table 7 Distributions of total sizes of websites

Total page size (MB)	Number of websites		
	Europe (n = 273)	North America (n = 20)	Oceania (n = 20)
0–1.4 (very good)	122 (%44.69)	6 (%30)	4 (%20)
1.5–2.4 (good)	65 (%23.81)	7 (%35)	7 (%35)
2.5–3.4 (average)	28 (%10.26)	3 (%15)	6 (%30)
3.5–4.4 (bad)	22 (%8.06)	2 (%10)	1 (%5)
4.5 and above (very bad)	36 (%13.19)	2 (%10)	2 (%10)

Table 8 Ranking of countries according to the number of websites in the "very good" and "very bad" ranges

Country	Continent	Number of websites
0–1.4 MB (very good) range		
Germany	Europe	8
Italy	Europe	7
Romania	Europe	7
Austria	Europe	6
Bulgaria	Europe	6
Norway	Europe	6
4.5 MB and above (very bad) range		
Belarus	Europe	3
Denmark	Europe	2
Belgium	Europe	2
Poland	Europe	2
Finland	Europe	2
UK	Europe	2

level. As a result of the analysis, a website with the minimum risk level was not found. In this context, the percentage distribution of the risk levels of the sites to the continents is shown in Fig. 3. The distribution of risk levels across all continents showed a majority at low and medium levels. The majority of prospective student web pages on the Oceanian continent were found to be at low risk levels. Critical risk web pages comprised less than 1% of all websites, and these websites belonged to the European continent. Overall, the analysis results showed that Oceania had the best performance, followed by Europe and North America with similar performance.

Table 9 presents the general distribution of web servers. The distribution of web servers across all continents showed that the most frequently used web server was Apache with almost 50%. This rate was followed by the Nginx web server with approximately one-third usage rate. The distribution of web server types by continents is given in Fig. 4. Likewise, Apache was the most frequently used web server on almost all continents, followed by Nginx.

Additionally, a quarter of the websites in North America and Oceania and about 40% of the websites in Europe were publicly publishing the version information of their web servers. When the websites for which the version information was accessed were examined, the problem of using outdated software on the web servers was detected. As a result of the analysis, 20% of the servers had old versions.

In the last phase of the security analysis, the use of TLS (SSL) encryption on websites was investigated, and the results revealed that 25 out of 330 websites do not use SSL encryption.

Table 9 Distribution of web servers

Type of web server	Number of websites
Apache	152 (46.06%)
Nginx	86 (26.06%)
Microsoft-IIS	22 (6.67%)
Other	29 (8.79%)
Unknown	41 (12.42%)

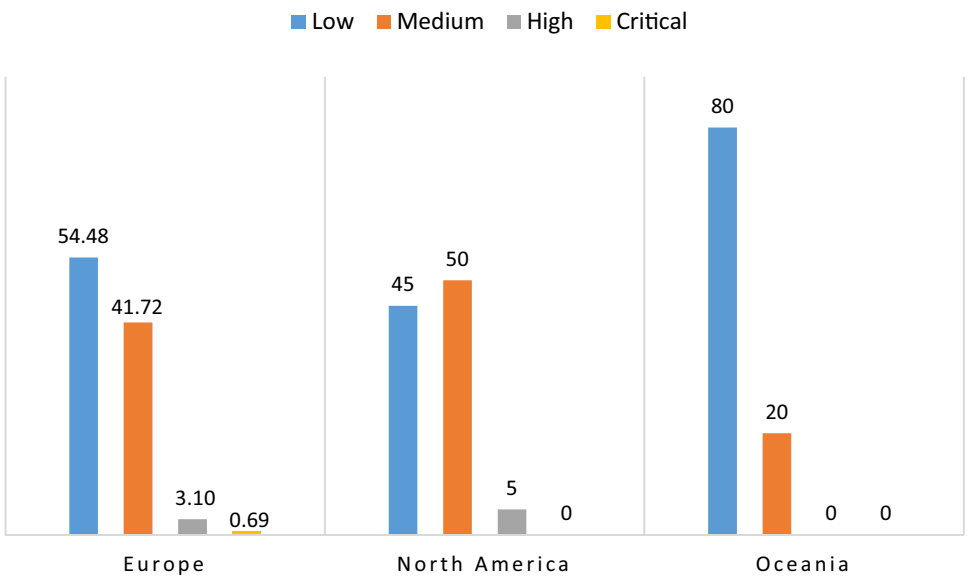


Fig. 3 Percentage of risk levels of the sites by continent

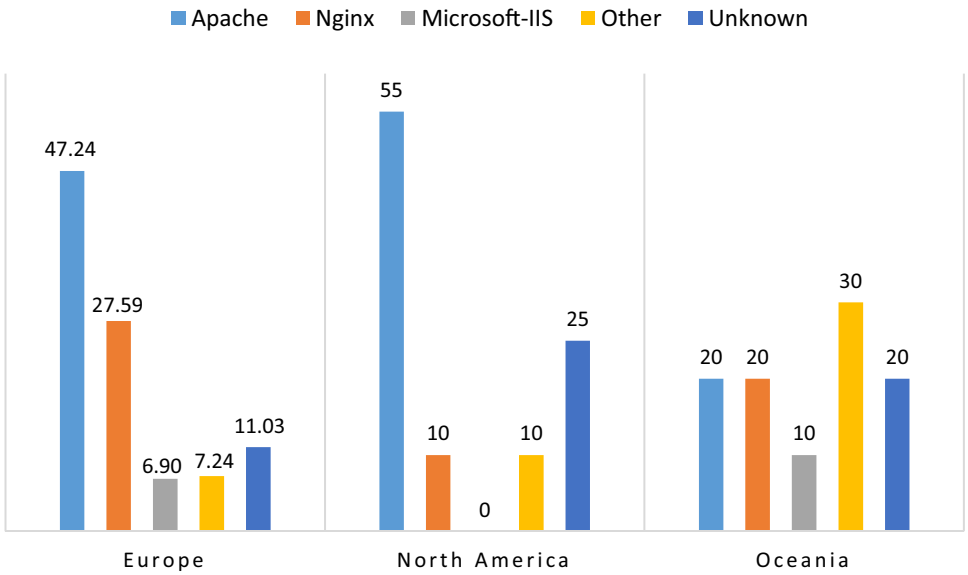


Fig. 4 Percentage of web servers by continent

Table 10 Continent rankings in all assessments

Assessment	Europe	North America	Oceania
Web accessibility			
Conformance level AA	2	1	3
Web usability			
Loading time	3	1	2
Page size	1	2	3
Web security			
Risk analysis	3	2	1

5 Discussion and conclusion

In this study, we examined the prospective student web pages of the leading universities in three continents, namely Europe, North America, and Oceania, in terms of accessibility, security and quality of use. According to the results of the analysis, the performances of the websites in three continents, Europe, North America, and Oceania, are ranked, and these rankings are presented in Table 10.

Based on the rankings in Table 10, we can deduce that university websites that rank higher in terms of web presence have less accessibility problems, the desired information can be accessed in a shorter time, and fewer security problems are detected. Therefore, it can be concluded that there is a relationship between web presence and web accessibility, usability, and security.

For the accessibility evaluation, 330 selected prospective student web pages were evaluated using the automated tool TAW, based on the WCAG 2.0 guidelines. The results showed that prospective student web pages had critical accessibility issues. For this reason, it can be concluded that prospective student web pages cannot be used effectively and satisfactorily by all stakeholders, especially by disabled prospective students. The TAW tool did not detect an error for conformance level AA, but results showed that none of the websites evaluated met the minimum conformance level A. In the comparison of the number of errors per web page, web pages in the continent of North America performed best, while the continent of Oceania performed worst. The results of this study determined that the most frequently violated check points at compliance level A were 1.1.1—Non-text Content and 4.1.1—Parsing. The results show similar performance to previous research studies [7, 9, 28, 70] in web accessibility analysis of university homepages. In short, the results of the analysis show that similar accessibility errors on the university home pages are also present on the web pages for prospective students. For this reason, we recommend that developers and administrators pay attention to guidelines such as WCAG 2.0 when designing websites with a large number of stakeholders, such as university and prospective student websites.

In terms of usability, the performance scores, loading times, and page sizes of the web pages for prospective students were examined. Page performance control results show that more than half of the websites have performance values below the acceptable level. In an overview of the websites, countries in North America had the best performance in the continent and cross-country rankings, followed by Oceanian and European countries. Eighty percentage of websites in North America had an acceptable performance grade. Almost two-thirds of European websites had below-average performance values. Additionally, increased page load times are associated with HTTP request counts and page sizes. The

overall page size analysis results showed that about one-third of websites exceeded the average level. We offer suggestions on optimizing image sizes, reducing the number of queries, and reducing the size of the website. With the use of web optimization applications, developers will increase the performance of the pages by combining or minifying HTML, CSS, and JavaScript content.

Within the scope of security analysis of web pages, among the websites evaluated, Oceania showed the best performance, followed by North America and Europe. The analysis of SSL encryption results showed that more than 90% of websites provide secure communication. More than a third of the websites evaluated publicly publish the version information of their web servers, increasing the risk of websites being hacked. Prospective student pages in the European continent constitute a large part of this rate. One should bear in mind that it is important to take such security measures in order to reduce the risk of attack and enable secure communication.

This analysis study was performed using online automated testing tools. Therefore, the limitation of the study is that the evaluation was performed using only automated test tools. Within the scope of future studies, especially within the scope of accessibility and usability criteria, the missing aspect will be completed by focusing on the user experience.

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